

Harmonic oscillator 1

Let $V(x)$ be the potential energy for a harmonic oscillator.

$$V(x) = \frac{m\omega^2 x^2}{2}$$

We seek $\psi(x, t)$ that solves the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + V(x) \psi(x, t)$$

The solution is

$$\psi_n(x, t) = C_n \exp \left[\frac{m\omega x^2}{2\hbar} - i\omega t \left(n + \frac{1}{2} \right) \right] \frac{d^n}{dx^n} \exp \left(-\frac{m\omega x^2}{\hbar} \right)$$

where $n = 0, 1, 2, \dots$ and C_n is the normalization constant

$$C_n = \frac{(-1)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} \left(\frac{\hbar}{m\omega} \right)^{n/2}$$

Evaluate the time derivative of $\psi(x, t)$ to obtain

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = \hbar\omega \left(n + \frac{1}{2} \right) \psi(x, t)$$

Hence $\psi(x, t)$ also solves the time-independent Schrodinger equation

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + V(x) \psi(x, t) = E_n \psi(x, t) \quad (1)$$

where

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right)$$

The exponential of t cancels in (1) hence $\psi_n(x)$ is also a solution where

$$\psi_n(x) = C_n \exp \left(\frac{m\omega x^2}{2\hbar} \right) \frac{d^n}{dx^n} \exp \left(-\frac{m\omega x^2}{\hbar} \right)$$

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